OmniLaTeX Press

OmniLaTeX

A Modular Document Class for Academic and Professional Documents

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1 Introduction

OmniLaTeX is a modular LaTeX document class built on LuaLaTeX and KOMA-Script. It provides 46 doctype aliases, 21 modular packages, and support for academic and professional documents including theses, articles, CVs, technical reports, patents, and more.

1.1 Requirements

- LuaLaTeX (TeX Live 2024 or newer)
- KOMA-Script (bundled with TeX Live)
- polyglossia (for multilingual support)

1.2 Installation

1.2.1 From CTAN (recommended)

Once published on CTAN, install via:

```
tlmgr install omnilatex
```

1.2.2 From Source

Clone the repository and ensure the `lib/` and `config/` directories are in your `TEXINPUTS` path.

1.3 Quick Start

```latex
\documentclass[doctype=article]{omnilatex}
\begin{document}
\title{My Document}
\author{Author Name}
\maketitle
\tableofcontents
\section{Introduction}
Hello, world!
\end{document}
```

Compile with:

```
latexmk -lualatex main.tex
```

2 Class Options

OmniLaTeX accepts the following options via `\documentclass`:

Option	Description
`language`	Document language (default: `english`). Passed to polyglossia.
`doctype`	Document type alias (default: `thesis`). See Chapter 3.
`titlestyle`	Title page style (default: `book`).
`institution`	Institution configuration (default: `none`). See Chapter 4.
`censoring`	Enable redaction support (default: `false`).
`loadGlossaries`	Enable glossaries module (default: `false`).
`todonotes`	Enable TODO notes (default: `false`).
`enablefonts`	Enable custom font loading (default: `false`).
`enablegraphics`	Enable graphics module (default: `false`).
`enablemath`	Enable math module (default: `true`).
`enabletikz`	Enable TikZ modules (default: `true`).
`enableengineering`	Enable engineering TikZ shapes (default: `true`).

`enablecode`	Enable code listings (default: `true`).
`enabletables`	Enable table module (default: `true`).

3 Document Types

OmniLaTeX provides 46 doctype aliases that resolve to 13 document profiles across 3 KOMA-Script base classes.

3.1 Available Doctypes

Alias	Profile	Base Class
thesis	Thesis	scrbook
dissertation	Dissertation	scrbook
book	Book	scrbook
manual	Manual	scrbook
article	Article	scrartcl
journal	Journal	scrartcl
cv	CV	scrartcl
cover-letter	Cover Letter	scrartcl
technicalreport	Technical Report	scrreprt
standard	Standard	scrreprt
patent	Patent	scrreprt

dictionary	Dictionary	scrbook
inlinepaper	Inline Paper	scrartcl

Additional aliases exist for common variations (e.g., `report`, `paper`, `resume`, `letter`). See `omnilatex.cls` for the complete alias table.

3.2 Doctype Profiles

Each doctype profile (in `config/document-types/`) defines:

- Page layout (`DIV`, margins, headers/footers)
- Font size
- Line spacing
- Title page style
- Section numbering depth
- Table of contents depth
- Color mode (color, grayscale, bw)

4 Institution Configuration

OmniLaTeX supports pluggable institution branding via `config/institutions/`.

4.1 Using an Institution

```
\documentclass[doctype=thesis,institution=tuhh]{omnilatex}
```

This loads `config/institutions/tuhh/tuhh.sty`, which provides:

- Logo definitions (language-aware via `\DeclareTranslation`)
- Institution links
- Color palette
- Custom commands (degree names, department names, etc.)

4.2 Creating a Custom Institution

See `CONTRIBUTING.md` for a step-by-step guide.

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5 Modules

OmniLaTeX is organized into 21 modules across 9 subdirectories:

5.1 Core (`lib/core/`)

`omnilatex-base` Build mode detection (dev/prod/ultra), date/time utilities.

5.2 Layout (`lib/layout/`)

`omnilatex-page` Headers, footers, page makeup via scrlayer-scrpage.

`omnilatex-floats` Float placement, captions, subfloats, source notation.

`omnilatex-document` Document metadata, custom `\document*` commands.

`omnilatex-boxes` Color boxes via tcolorbox.

`omnilatex-koma` KOMA-Script sectioning, titlepage, TOC customization.

5.3 Typography (`lib/typography/`)

`omnilatex-fonts` Font loading with graceful fallbacks.

`omnilatex-typesetting` Line spacing, dashes, quotation marks.

`omnilatex-math` Math typesetting, delimiters, physics macros, big-O notation.

`omnilatex-lists` Custom list environments via enumitem.

5.4 References (`lib/references/`)

`omnilatex-hyperref` Hyperlinks, PDF metadata, git SHA embedding.

`omnilatex-biblio` Bibliography via biblatex (biber backend).

`omnilatex-glossary` Glossaries, abbreviations, symbols via bib2gls.

5.5 Language (`lib/language/`)

`omnilatex-i18n` Polyglossia integration, translations.

5.6 Graphics (`lib/graphics/`)

`omnilatex-graphics` Image handling, SVG support (Inkscape).

`omnilatex-tikz-core` TikZ/PGFPlots core, forest trees.

`omnilatex-tikz-engineering` Engineering diagrams, P&ID, flowcharts.

5.7 Code (`lib/code/`)

`omnilatex-listings` Source code highlighting via minted.

5.8 Tables (`lib/tables/`)

`omnilatex-tables` Table formatting (booktabs, tabularray, longtable).

5.9 Utilities (`lib/utils/`)

`omnilatex-utils` TODO notes, git metadata, keyboard macros, censoring.

`omnilatex-colors` Color definitions and schemes.

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6 Font System

OmniLaTeX uses a three-font stack:

Libertinus Serif + Math Main text and mathematics. Always available via TeX Live.

Monaspace Neon Monospace and code listings. Optional — falls back to Latin Modern Mono.

Atkinson Hyperlegible Next Sans-serif elements. Optional — falls back to Libertinus Sans.

Missing optional fonts trigger a `\ClassWarning` and degrade gracefully. No manual configuration is needed.

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7 Build Modes

OmniLaTeX supports three build modes, controlled via the `OMNILATEX_MODE` environment variable:

dev Development mode. Code listings use basic formatting, debug output enabled.

prod Production mode (default). Full formatting, no debug output.

ultra Ultra mode. Maximum quality, SVG conversion, no compromises.

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8 Reproducible Builds

OmniLaTeX supports deterministic PDF generation via `SOURCE_DATE_EPOCH`:

```
SOURCE_DATE_EPOCH=$(git log -1 --format=%ct) latexmk -lualatex main.tex
```

This sets all PDF timestamps to the git commit date, enabling byte-for-byte reproducible builds.

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9 License

OmniLaTeX is licensed under the Apache License 2.0.